

RISK & RETURN: THE MATHEMATICS OF INVESTING

Financial Mathematics (TE7677) · CA-III Poster · Unit V: Basics of Investment & Unit VI: Risk and Uncertainty

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UNIT V: BASICS OF INVESTMENT

1. Investment Return

Return = the gain (or loss) on an investment as a % of the money put in.

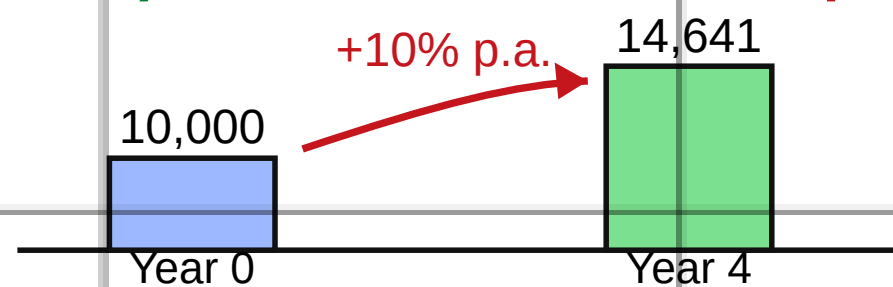
$$R = (V_1 - V_0 + \text{Income}) / V_0 \times 100$$

For many years use the **CAGR** (average yearly growth):

$$\text{CAGR} = (V_n / V_0)^{1/n} - 1$$

Eg: ₹10,000 grows to ₹14,641 in 4 yrs

$$\text{CAGR} = (1.4641)^{1/4} - 1 = 1.10 - 1 = 10\% \text{ p.a.}$$



2. Compounding Frequency of Interest

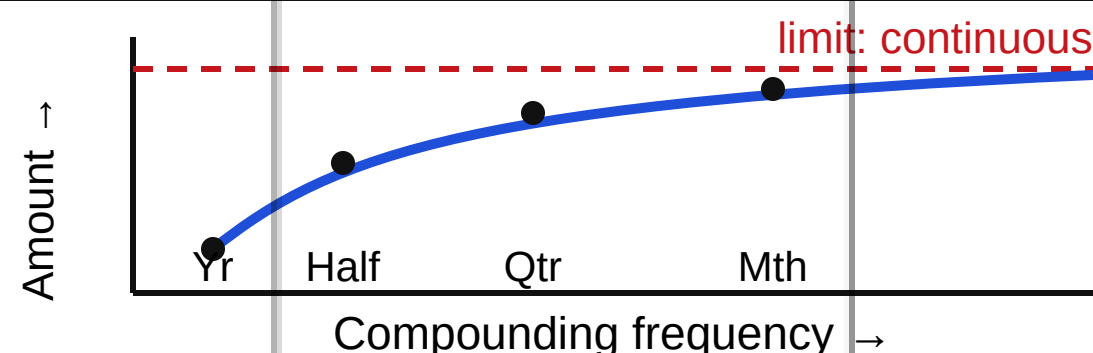
More frequent compounding $\Rightarrow$ more interest. Nominal rate r , m periods/yr, t years:

$$A = P(1 + r/m)^{mt} \quad | \quad \text{Continuous: } A = P e^{rt}$$

$$\text{Effective rate: } i_{\text{eff}} = (1 + r/m)^m - 1 \quad (\text{cont.: } e^r - 1)$$

Eg: $P = ₹10,000$, $r = 12\% \text{ p.a.}$, $t = 1 \text{ yr}$

Compounding	m	Amount A (₹)	Eff. rate
Annual	1	11,200.00	12.00%
Half-yearly	2	11,236.00	12.36%
Quarterly	4	11,255.09	12.55%
Monthly	12	11,268.25	12.68%
Continuous	∞	11,274.97	12.75%



3. Uneven Cash Flows

Cash flows that are **not equal** every period. Discount each one separately:

$$PV = \sum CF_t / (1 + i)^t \quad \text{NPV} = PV - \text{Initial cost}$$

Eg: invest ₹4,500 now. Receive ₹1,000, ₹2,000, ₹3,000 at end of yrs 1, 2, 3. $i = 10\%$.



LINK: Higher risk $\Rightarrow$ higher required return (Unit V gives the return, Unit VI prices the risk). Good investing = best return for the risk you can bear.

Yr	CF	Divide by	PV
1	1,000	1.10	909.09
2	2,000	1.21	1,652.89
3	3,000	1.331	2,253.94
Total PV ($\approx$)			₹4,816

$$\text{NPV} \approx 4,816 - 4,500 = +₹316 > 0 \Rightarrow \text{ACCEPT}$$

4. Economic Equivalence

Different amounts at different times are **equivalent** if they have the **same value** at one common date, at the given interest rate. (₹1 today $\neq$ ₹1 tomorrow.)

$$F = P(1 + i)^n \quad \Leftrightarrow \quad P = F / (1 + i)^n$$

Eg: at $i = 10\% \text{ p.a.}$, all three below are equivalent:

$$₹9,091 \quad ₹10,000 \quad ₹12,100$$

$$10,000 \times (1.10)^2 = 12,100 \quad | \quad 10,000 / 1.10 = 9,090.91$$

Use: compare loans, pick between ₹10,000 today vs ₹12,500 in 2 yrs ($12,100 < 12,500$, so wait!).

UNIT VI: RISK AND UNCERTAINTY

1. Decision under Risk and Uncertainty

Risk: outcomes AND their probabilities are known $\rightarrow$ use

EMV = $\sum p \times \text{payoff}$.

Uncertainty: probabilities unknown $\rightarrow$ use decision rules.

Eg: Profit (₹ thousand) of 3 investments. $P(\text{Boom, Normal, Recession}) = 0.3, 0.5, 0.2$

Option	Boom 0.3	Normal 0.5	Recession 0.2	EMV
A: Stocks	80	40	-20	40.0
B: Bonds	40	30	20	31.0
C: Real estate	50	35	0	32.5

$$\text{EMV}(A) = 0.3(80) + 0.5(40) + 0.2(-20) = 24 + 20 - 4 = 40$$

Rules when probabilities are unknown (same table):

Rule	Idea	Pick
Maximax	optimist: best of best (80, 40, 50)	A
Maximin	pessimist: best of worst (-20, 20, 0)	B
Min-max regret	least max regret (40, 40, 30)	C
Laplace	equal chances: avg (33.3, 30, 28.3)	A

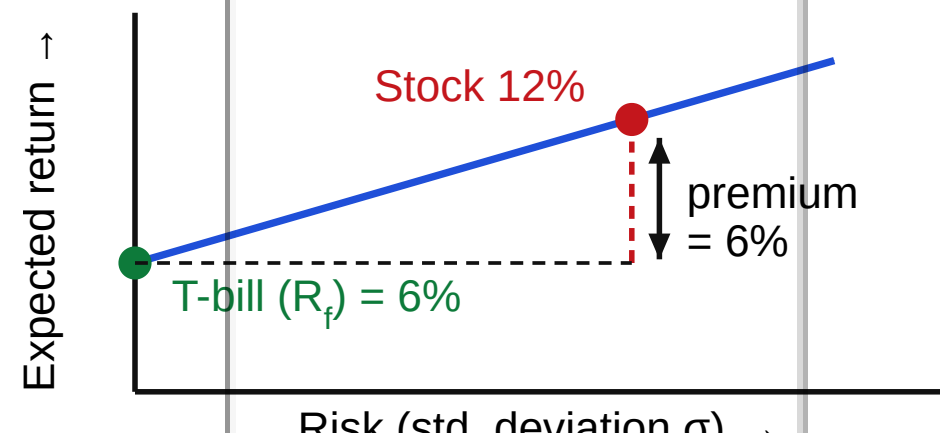
Regret = best payoff in that state – your payoff. Different rules $\Rightarrow$ different answers, so the investor's attitude to risk matters.

2. Risk Premium

Extra return demanded for taking risk instead of a safe asset.

$$\text{Risk premium} = E(R) - R_f$$

Eg: Stock $E(R) = 12\%$, T-bill $R_f = 6\% \Rightarrow$ premium = 6%



Required return = Risk-free rate + Risk premium

3. Portfolio Diversification

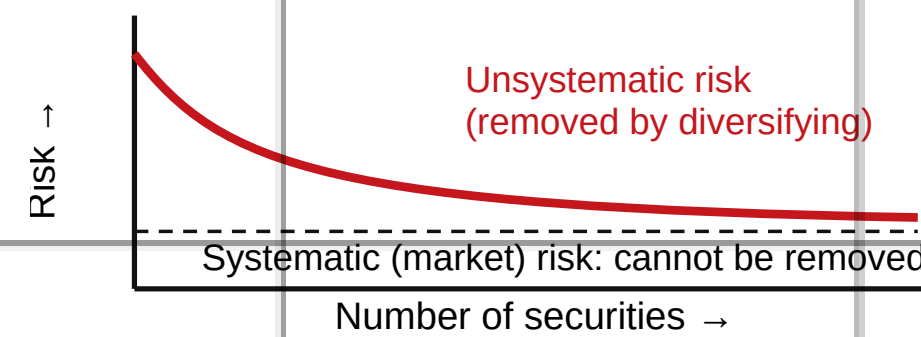
"Don't put all eggs in one basket": spreading money cuts risk without cutting average return.

$$E(R_p) = w E(R_1) + (1-w) E(R_2)$$

$$\sigma_p^2 = w^2 \sigma_1^2 + (1-w)^2 \sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$$

Eg: two stocks, $\sigma = 20\%$ each, $w = 50\%$:

Correlation ρ	+1	+0.5	0	-1
Portfolio σ	20%	17.3%	14.1%	0%



4. Life Insurance

A contract: you pay a **premium**, the insurer pays the **sum**

the many (law of large numbers).

$$\text{Net single premium (1-yr term)} = S \times q / (1 + i)$$

Eg: $S = ₹1,00,000$, $q = P(\text{death in the year}) = 0.01$, $i = 5\%$

$$\text{NSP} = 1,00,000 \times 0.01 / 1.05 = ₹952.38$$

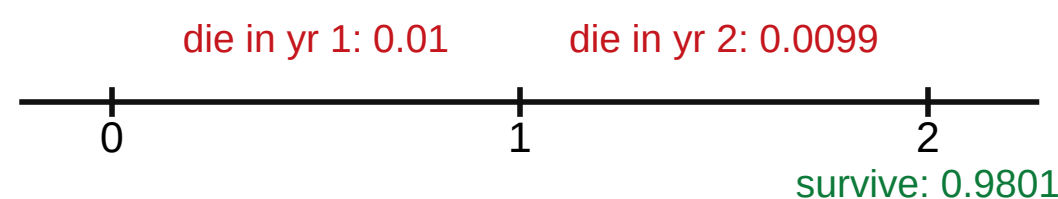
Types: **Term** (death only), **Whole life** (death, any time), **Endowment** (death or survival). Actual premium = NSP + expenses + profit.

5. Endowment Assurance

Pays the sum assured if you die within the term OR

(survival benefit)

Eg: 2-yr plan, $S = ₹1,00,000$, $q = 0.01$ each yr, $i = 5\%$



$$\text{NSP} = 1,00,000 [0.01/1.05 + 0.0099/1.05^2 + 0.9801/1.05^2] = 1,850 + 88,898 = ₹90,748 \text{ (mostly the savings part)}$$

Reference: Campolieti & Makarov, *Financial Mathematics: A Comprehensive Treatment*; Capinski & Zastawniak, *Mathematics for Finance*; Hull & Basu, *Options, Futures & Other Derivatives* (Pearson).